

概 率 论



数学期望

$$\begin{aligned} ① \quad & \begin{cases} X \sim p_i & EX = \sum_{i=1}^{\infty} x_i p_i \\ Y = g(X) & EY = \sum_{i=1}^{\infty} g(x_i) \cdot p_i \end{cases} \\ ② \quad & \begin{cases} X \sim f(x) & EX = \int_{-\infty}^{+\infty} x f(x) dx \\ Y = g(X) & EY = \int_{-\infty}^{+\infty} g(x) f(x) dx \end{cases} \\ ③ \quad & \begin{cases} \text{线性: } E(\sum_{i=1}^n a_i X_i) = \sum_{i=1}^n a_i EX_i \\ E(aX+c) = aEX+c \\ E(X \pm Y) = EX \pm EY \end{cases} \end{aligned}$$

$$\begin{aligned} X, Y \text{ 独立: } & E(XY) = E(X)E(Y) \\ & E(g_1(X)g_2(Y)) = E(g_1(X))E(g_2(Y)) \end{aligned}$$

方差

$$\begin{aligned} DX &= E(X-EX)^2 = E(X^2 - 2X \cdot EX + (EX)^2) \\ &= EX^2 - 2(EX)^2 + (EX)^2 \\ &= EX^2 - (EX)^2 \end{aligned}$$

$$\begin{aligned} \text{标准化} \quad & \frac{X-EX}{\sqrt{DX}} \\ DX \geq 0, \quad & E(X^2) = DX + (EX)^2 \\ & E(X^2) \geq (EX)^2 \end{aligned}$$

$$Dc = 0$$

$$D(aX+b) = a^2 DX$$

$$D(X \pm Y) = DX \pm DY \pm 2\text{Cov}(X, Y)$$

$$X, Y \text{ 独立: } a^2 DX + b^2 DY$$

0-1 分布	$P\{X=k\} = p^k(1-p)^{1-k}, k=0,1$	p	$p(1-p)$
= 二项分布	$P\{X=k\} = C_n^k p^k (1-p)^{n-k}, k=0,1,\dots,n$	np	$np(1-p)$
泊松分布	$P\{X=k\} = \frac{\lambda^k}{k!} e^{-\lambda}, k=0,1,\dots,n$	λ	λ
几何分布	$P\{X=k\} = (1-p)^{k-1} p, k=1,2,\dots,n$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
正态分布	$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < +\infty$	μ	σ^2
均匀分布	$f(x) = \frac{1}{b-a}, a < x < b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
指数分布	$f(x) = \lambda e^{-\lambda x}, x > 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

切比雪夫不等式 独立与不相关

$$P\{|X-EX| \geq \varepsilon\} \leq \frac{DX}{\varepsilon^2} \quad P\{|X-EX| < \varepsilon\} \geq 1 - \frac{DX}{\varepsilon^2}$$

独立通过分布来判定，相关通过数字特征判定

$$\begin{aligned} \text{独立} \Rightarrow & \rho_{XY} = 0 \Leftrightarrow \text{Cov}(X, Y) = 0 \Leftrightarrow E(XY) = E(X)E(Y) \Leftrightarrow D(X \pm Y) = DX + DY \\ & \leftarrow \text{二维正态可以推回去} \end{aligned}$$

Cov

$$\text{Cov}(X, Y) = E[(X-EX)(Y-EY)] = E(XY) - E(X)E(Y)$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{DX} \sqrt{DY}}$$

$$E(XY) = \begin{cases} \sum_i \sum_j x_i y_j P\{X=x_i, Y=y_j\} & (\text{离散}) \\ \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f(x, y) dx dy & (\text{连续}) \end{cases}$$

$$\text{对称性 } \text{Cov}(X, Y) = \text{Cov}(Y, X), \quad \rho_{XY} = \rho_{YX}$$

$$\text{Cov}(X, X) = DX, \quad \rho_{XX} = 1$$

$$\text{线性性 } \text{Cov}(X, c) = 0, \quad \text{Cov}(aX+b, Y) = a\text{Cov}(X, Y)$$

$$\text{Cov}(X_1+X_2, Y) = \text{Cov}(X_1, Y) + \text{Cov}(X_2, Y)$$

= 二维正态 联合，
的条件，边缘 线性组合也是正态。

例4.4

例 4.1 设随机变量 X 的分布函数为

$$F(x) = \begin{cases} 0, & x < -1, \\ 0.2, & -1 \leq x < 0, \\ 0.8, & 0 \leq x < 1, \\ 1, & x \geq 1, \end{cases}$$

则 $EX = \underline{\hspace{2cm}}$, $E(2X+5) = \underline{\hspace{2cm}}$, $E(X^2) = \underline{\hspace{2cm}}$, $D(X^2) = \underline{\hspace{2cm}}$.

$$X \sim \begin{pmatrix} -1 & 0 & 1 \\ 0.2 & 0.6 & 0.2 \end{pmatrix}$$

$$EX = \sum_{i=1}^n X_i P_i = 0.2 \times (-1) + 0.6 \times 0 + 0.2 \times 1 = 0$$

$$EX^2 = 0.2 \times (-1)^2 + 0.6 \times 0^2 + 0.2 \times 1^2 = 0.4$$



例 4.8 设连续型随机变量 X 服从参数为 1 的指数分布, 则 $E(X + e^{-2X}) = \underline{\hspace{2cm}}$, $D(e^{-2X}) = \underline{\hspace{2cm}}$.

$$E(e^{-2X}) = \int_0^{+\infty} e^{-2x} \cdot \frac{e^{-x}}{1 \cdot x^1} dx = \int_0^{+\infty} e^{-3x} dx = \frac{1}{3}$$

$$D(e^{-2X}) = E(e^{-4X}) - \frac{1}{9} = -\frac{1}{9} + \int_0^{+\infty} e^{-4x} \cdot e^{-x} dx = -\frac{1}{9} + \frac{1}{5} = \frac{4}{45}$$

$$\begin{aligned} E(X + e^{-2X}) &= E(X) + E(e^{-2X}) \\ &= 1 + \frac{1}{3} = \frac{4}{3} \end{aligned}$$

例4.9

例 4.9 设连续型随机变量 X 的概率密度为

$$f(x) = \frac{1}{\pi(1+x^2)} \quad (-\infty < x < +\infty),$$

求 $E(\min\{|X|, 1\})$.

$$\begin{aligned} E(\min\{|X|, 1\}) &= \int_{-\infty}^{+\infty} \min\{|x|, 1\} \cdot f(x) dx = 2 \int_0^1 \frac{x}{\pi(1+x^2)} dx + 2 \int_1^{+\infty} \frac{1}{\pi(1+x^2)} dx \\ &= \frac{1}{\pi} \int_0^1 \frac{dx}{1+x^2} + \frac{2}{\pi} \int_1^{+\infty} \frac{1}{1+x^2} dx \\ &= \frac{1}{\pi} (\arctan x) \Big|_0^1 + \frac{2}{\pi} (\frac{\pi}{2} - \arctan x) \Big|_1^{+\infty} \\ &= \frac{\ln^2}{\pi} + \frac{2}{\pi} (\frac{\pi}{2} - \frac{\pi}{4}) = \frac{\ln^2}{\pi} + \frac{1}{2} \end{aligned}$$

例4.7

例 4.7 设连续型随机变量 X 的概率密度为

$$f(x) = \frac{1}{\sqrt{\pi}} e^{-x^2+2x-1} \quad (-\infty < x < +\infty),$$

则 $EX = \underline{1}$, $\sqrt{DX} = \underline{\frac{1}{2}}$.

$$f(x) = \frac{1}{\sqrt{\pi}} \frac{1}{\sqrt{\pi}} e^{-\frac{(x-1)^2}{2 \cdot \frac{1}{2}}} \Rightarrow X \sim N(1, \frac{1}{2})$$

例4.6

例 4.6 设连续型随机变量 X 的概率密度为

$$f(x) = \frac{1}{\pi(1+x^2)} \quad (-\infty < x < +\infty),$$

则 $EX(\quad)$.

(A) 等于 0

(B) 等于 1

(C) 等于 π

(D) 不存在

例4.11

设随机变量 X 与 Y 的概率分布分别为

X	0	1
P	$\frac{1}{3}$	$\frac{2}{3}$

Y	-1	0	1
P	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$

$X \backslash Y$	$-\frac{1}{3}$	0	$\frac{1}{3}$
$\frac{1}{3}$	0	$\frac{1}{3}$	0
$\frac{2}{3}$	$\frac{1}{3}$	0	$\frac{1}{3}$

$$Z \sim \begin{pmatrix} 0 & -1 & 1 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

$$\rho_{xy} = \frac{\text{Cov}(X, Y)}{\sqrt{D_X} \sqrt{D_Y}} = \frac{E(XY) - E(X)E(Y)}{\sqrt{D_X} \sqrt{D_Y}} = 0$$

$$P\{X+Y=0\}$$



且 $P\{X^2=Y^2\}=1$.

(1) 求随机变量 (X, Y) 的概率分布;

(2) 求 $Z=XY$ 的概率分布;

(3) 求 X 与 Y 的相关系数 ρ_{xy} .

例4.14



设随机变量 X 的概率密度为

$$f(x) = \frac{1}{2} e^{-|x|}, x \in (-\infty, +\infty).$$

证明: (1) X 与 $|X|$ 不相关; (2) X 与 $|X|$ 不独立.

$$\begin{aligned} (1) \quad \text{Cov}(X, |X|) &= E(X|X|) - E(X)E(|X|) \\ &= \int_{-\infty}^{+\infty} \frac{x}{2} e^{-|x|} dx - \left(\int_{-\infty}^{+\infty} \frac{x}{2} e^{-|x|} dx \right) \left(\int_{-\infty}^{+\infty} \frac{|x|}{2} e^{-|x|} dx \right) \\ &= \int_{-\infty}^{+\infty} \frac{x}{2} e^{-x^2} dx = 0 \Rightarrow \text{Cov}(X, |X|) = 0 \end{aligned}$$

$$(2) \quad P(A) = P\{X \leq \frac{1}{2}\} \neq 1$$

$$P(B) = P\{|X| \leq \frac{1}{2}\}$$

$$P(AB) = P\{|X| \leq \frac{1}{2}\} = P(B) \neq P(A)P(B)$$

例4.15

设 X, Y 为随机变量, 数学期望都是 2, 方差分别为 1 和 4, 相关系数为 0.5, 试用切比雪夫不等式估计概率 $P\{|X-Y| \geq 6\}$.

$$\begin{aligned} P\{|X-Y| \geq 6\} &\leq \frac{D(X-Y)}{\varepsilon^2} \\ &= \frac{D(X)+D(Y)-2\text{Cov}(X,Y)}{\varepsilon^2} \\ &= \frac{1+4-2}{36} = \frac{1}{12} \end{aligned}$$

$\text{Cov} = \rho \cdot \sqrt{D_X} \sqrt{D_Y} = 0.5 \times 1 \times 2$

习4.1

4.1 已知连续型随机变量 X 与 Y 有相同的概率密度, 且

$$X \sim f(x) = \begin{cases} 2x\theta^2, & 0 < x < \frac{1}{\theta}, \\ 0, & \text{其他} \end{cases} \quad (\theta > 0), \quad E[a(X+2Y)] = \frac{1}{\theta},$$

则 $a =$ (B).

(A) $\frac{2}{3}$

(B) $\frac{1}{2}$

(C) $\frac{1}{3}$

(D) $\frac{1}{6}$

$$EX = \int_0^{\frac{1}{\theta}} 2x^2 \theta^2 dx = \frac{2\theta^2}{3} x^3 \Big|_0^{\frac{1}{\theta}} = \frac{2}{3\theta}$$

$$E(a(X+2Y)) = \frac{2a}{\theta} = \frac{1}{\theta} \quad a = \frac{1}{2}$$

习4.2

4.2 已知连续型随机变量 Y 的概率密度为

$$f(y) = \begin{cases} \frac{y}{a^2} e^{-\frac{y}{a}}, & y > 0, \\ 0, & y \leq 0, \end{cases} \quad a > 0,$$

求随机变量 $Z = \frac{1}{Y}$ 的数学期望 EZ .

$$\begin{aligned} EZ &= \int_0^{+\infty} \frac{1}{y} \cdot \frac{y}{a^2} e^{-\frac{y}{a}} dy \\ &= \frac{1}{a} \int_0^{+\infty} \frac{1}{ay} e^{-\frac{1}{2ay}} d ay \\ &= \frac{1}{a} \int_0^{+\infty} \frac{1}{t} e^{-\frac{1}{2t}} dt \\ &= -\frac{1}{a} \int_0^{+\infty} e^{-\frac{1}{2t}} d \frac{1}{t} \\ &= \frac{1}{a} \int_0^{+\infty} e^{-u^2} du \\ &= \frac{1}{a} \cdot \frac{\sqrt{\pi}}{2} \end{aligned}$$

习4.3

4.3 甲、乙两人相约于某地在 12:00~13:00 会面, 设 X, Y 分别是甲、乙到达超过 12:00 的时间 (单位: 小时), 且假设 X 和 Y 相互独立, 已知 X, Y 的概率密度分别为

$$f_X(x) = \begin{cases} 3x^2, & 0 < x < 1, \\ 0, & \text{其他}, \end{cases} \quad f_Y(y) = \begin{cases} 2y, & 0 < y < 1, \\ 0, & \text{其他}. \end{cases}$$

求先到达者需要等待的时间的数学期望.



$$\begin{aligned} E(|X-Y|) &= \iint_D |x-y| 3x^2 \cdot 2y \, d\sigma \\ &= \int_0^1 dx \int_0^x (x-y) 6x^2 y \, dy + \int_0^1 dx \int_y^1 (y-x) 6x^2 y \, dy \\ &= \int_0^1 3x^3 y^2 - 2x^2 y^3 \, dx + \int_0^1 (3x^3 y^2 - 2x^2 y^3) - (3x^3 - 2x^2) \, dx \\ &= \int_0^1 2x^5 - 3x^3 + 2x^2 \, dx \\ &= \frac{1}{3} x^6 - \frac{3}{4} x^4 + \frac{2}{3} x^3 \Big|_0^1 \\ &= \frac{1}{3} - \frac{3}{4} + \frac{2}{3} = \frac{1}{4} \end{aligned}$$

习4.4

4.4 设 A, B 为两个随机事件, 且 $P(A) = \frac{1}{4}, P(B|A) = \frac{1}{3}, P(A|B) = \frac{1}{2}$. 令

$$X = \begin{cases} 1, & A \text{ 发生}, \\ 0, & A \text{ 不发生}, \end{cases} \quad Y = \begin{cases} 1, & B \text{ 发生}, \\ 0, & B \text{ 不发生}, \end{cases}$$

求: (1) 二维随机变量 (X, Y) 的概率分布;

(2) X, Y 的相关系数 ρ_{XY} ;

(3) $Z = X^2 + Y^2$ 的概率分布.

(1) $P(B|A) = \frac{P(AB)}{P(A)} \Rightarrow P(AB) = \frac{1}{12} \Rightarrow P(B) = \frac{P(AB)}{P(A|B)} = \frac{\frac{1}{12}}{\frac{1}{2}} = \frac{1}{6}$

	Y	1	0	Σ
X	1	$\frac{1}{12}$	$\frac{1}{8}$	$\frac{5}{24}$
	0	$\frac{1}{12}$	$\frac{1}{8}$	$\frac{1}{6}$
Σ		$\frac{1}{6}$	$\frac{5}{12}$	1

(2) $\rho_{XY} = \frac{E(XY) - E(X)E(Y)}{\sqrt{D_X D_Y}} = \frac{\frac{1}{12} - \frac{1}{4} \cdot \frac{1}{6}}{\sqrt{\frac{5}{16} \cdot \frac{5}{36}}} = \frac{\sqrt{5}}{5}$

$$(3) Z \sim \begin{pmatrix} 0 & 1 & 2 \\ \frac{8}{12} & \frac{3}{6} & \frac{1}{6} \end{pmatrix}$$

4.5

4.5 设随机变量 (X, Y) 服从二维正态分布, X 与 Y 的期望值为 μ , 方差为 σ^2 , X, Y 的相关系数为 $\rho_{XY} = 0$, 记 $Z_1 = 2X + Y, Z_2 = 2X - Y$, 则 Z_1, Z_2 的相关系数为 (B).

- (A) 0 (B) $\frac{3}{5}$ (C) $\frac{3}{\sqrt{15}}$ (D) $\frac{3}{\sqrt{10}}$

$$\begin{aligned} & E[(2X+Y)(2X-Y)] \\ &= E[4X^2 - Y^2] \\ &= 4E[X^2] - E[Y^2] \\ &= 3[\sigma^2 + \mu^2] \end{aligned} \quad \left| \quad \begin{aligned} E[2X+Y] &= 3\mu & E[2X-Y] &= \mu \\ \Rightarrow E[2X+Y]E[2X-Y] &= 3\mu^2 \end{aligned} \right.$$

$$\Rightarrow \frac{E[(2X+Y)(2X-Y)] - E[2X+Y]E[2X-Y]}{\sqrt{\sigma^2} \cdot \sqrt{\sigma^2}} = \frac{3\sigma^2}{\sigma^2} = \frac{3}{5}$$

4.6

4.6 设 $X \sim N(0, \sigma^2) (\sigma > 0), Y = X^3$.

- (1) 求 EY ;
(2) 判断 X 与 Y 是否相关? 是否独立? 说明理由.

$$(1) \begin{cases} f_X = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{x^2}{2\sigma^2}} \\ f_Y = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{x^6}{2\sigma^6}} \end{cases} \rightarrow EY = 0$$

$$(2) \text{Cov}(X, Y) = \underbrace{E[X^4]}_{\neq 0} - \underbrace{E[X]}_{=0} \underbrace{E[Y]}_{=0} \neq 0 \text{ 相关} \rightarrow \text{不独立}$$

4.7

4.7 设随机变量 X 在 $[-1, 1]$ 上服从均匀分布, $Y = X^3$, 则 X 与 Y ().

- (A) 不相关且相互独立 (B) 不相关且相互不独立
(C) 相关且相互独立 (D) 相关且相互不独立

$$f_X = \begin{cases} \frac{1}{2}, & x \in (-1, 1) \\ 0, & \text{others} \end{cases}$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = E[X^4] \neq 0$$

$$F_X(-\frac{1}{2}) \neq F_X(\frac{1}{2}) \cdot F_Y(\frac{1}{8}) \quad \text{相关不独立}$$

4.8

4.8 设 X 为随机变量, $P\{|X - EX| < \varepsilon\} \geq 0.9, DX = 0.009$, 试用切比雪夫不等式估计 ε 的取值范围.

$$\begin{aligned} P\{|X - EX| < \varepsilon\} &\geq 1 - \frac{DX}{\varepsilon^2} \geq 0.9 \\ \Rightarrow \frac{DX}{\varepsilon^2} &\leq 0.1 \quad \varepsilon^2 \geq \frac{0.009}{0.1} = 0.09 \\ &\Rightarrow \varepsilon \geq 0.3 \end{aligned}$$

300.4.1

4.1 设连续型随机变量 X 的概率密度为

$$f(x) = \frac{2}{\pi(4+x^2)}, x \in (-\infty, +\infty),$$

则 EX (D).

- (A) 等于 0 (B) 等于 1 (C) 等于 π (D) 不存在

$$\int_{-\infty}^{+\infty} \frac{dx}{x^2+4} = \frac{2}{\pi} \ln(x^2+4) \Big|_{-\infty}^{+\infty}$$

300.4.2

4.2 已知随机变量 X, Y 相互独立, 且都服从泊松分布, 又如 $EX = 2, EY = 3$, 则 $E[(X+Y)^2] =$

- (A) 10 (B) 25 (C) 30 (D) 51

$$\begin{aligned} & E[X^2 + Y^2 + 2XY] \\ &= E[X^2] + E[Y^2] + 2E[XY] \\ &\quad \begin{matrix} 2+4 & 3+9 & 12 \end{matrix} \end{aligned}$$

300.4.3

4.3 已知连续型随机变量 X 与 Y 有相同的概率密度, 且
 $f(x) = \begin{cases} 3x^2\theta^3, & 0 < x \leq \frac{1}{\theta} \\ 0, & \text{其他} \end{cases} (\theta > 0)$,
 $E[X(X+2Y)] = \frac{1}{\theta}$, 则 $a =$ (C).

(A) $\frac{2}{9}$

(B) $\frac{1}{3}$

(C) $\frac{4}{9}$

(D) $\frac{2}{3}$

$$EX = \int_0^{\frac{1}{\theta}} 3x^2\theta^3 dx = \frac{3}{4}x^4\theta^3 \Big|_0^{\frac{1}{\theta}} = \frac{3}{4\theta}$$

$$E[(X+Y)^2] = \frac{9}{4\theta} = \frac{1}{\theta} \Rightarrow a = \frac{4}{9}$$

300.4.4

4.4 设随机变量 X 的概率密度为

$$f(x) = \begin{cases} kx^a, & 0 < x < 1, \\ 0, & \text{其他} \end{cases} (k, a > 0).$$

又知 $EX = \frac{3}{4}$, 则 k, a 分别为 _____.

$$EX = \int_0^1 kx^a dx = \frac{k}{a+1}x^{a+1} \Big|_0^1 = \frac{k}{a+1} = 1$$

$$= \frac{k}{a+1}x^{a+1} \Big|_0^1 = \frac{k}{a+1} = \frac{3}{4}$$

$$4a+4 = 3a+6 \rightarrow a=2 \quad k=3$$

300.4.5

4.5 设离散型随机变量 X 的分布函数为

$$F(x) = \begin{cases} 0, & x < -1, \\ 0.2, & -1 \leq x < 2, \\ 0.5, & 2 \leq x < 5, \\ 1, & x \geq 5, \end{cases}$$

则 $EX =$ _____.

$$EX = -1 \cdot 0.2 + 0.7 \cdot 2 + 0.5 \cdot 5$$

$$= -0.2 + 0.6 + 2.5 = 2.9$$

300.4.6

4.6 设随机变量 X 的概率密度为 $f(x) = \begin{cases} \frac{3}{8}x^2, & 0 < x < 2, \\ 0, & \text{其他} \end{cases}$ 则 $E(\frac{1}{X^2}) =$ _____.

$$E(\frac{1}{X^2}) = \int_0^2 \frac{1}{x^2} \cdot f(x) dx = \int_0^2 \frac{3}{8} dx = \frac{3}{4}$$

$$E[\frac{1}{X^2}] = \int_{\frac{1}{2}}^{+\infty} \frac{1}{x^2} \cdot \frac{3}{8x^2} dx = \int_{\frac{1}{2}}^{+\infty} \frac{3}{8} x^{-4} dx = -\frac{x^{-3}}{3} \Big|_{\frac{1}{2}}^{+\infty} = \frac{2\sqrt{2}}{8} = \frac{\sqrt{2}}{4}$$

$$0 < \frac{1}{X^2} < 2 \quad \frac{1}{2} < X^2$$

300.4.7

4.7 设相互独立的两个随机变量 X, Y 具有同一分布律, 且 X 的分布律为

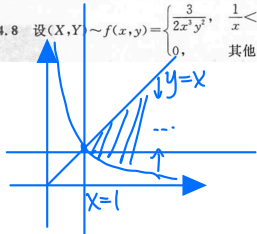
X	0	1
P	$\frac{1}{2}$	$\frac{1}{2}$

则随机变量 $Z = \max\{X, Y\}$ 的数学期望 $EZ =$ $\frac{3}{4}$.

$$Z \sim \begin{pmatrix} \frac{1}{4} & \frac{1}{2} \\ 0 & \frac{3}{4} \end{pmatrix}$$

300.4.8

4.8 设 $(X, Y) \sim f(x, y) = \begin{cases} \frac{3}{2x^3y^2}, & \frac{1}{x} < y < x, x > 1, \\ 0, & \text{其他} \end{cases}$ 则 $EY =$ _____, $E(\frac{1}{XY}) =$ _____.



$$f_Y(y) = \begin{cases} \int_1^{+\infty} \frac{3}{2x^3y^2} dx = \frac{3}{4}y & y \in (0, 1) \\ \int_y^{+\infty} \frac{3}{2x^3y^2} dx = \frac{3}{2y^2} & y \in (1, +\infty) \end{cases}$$

$$EY = \int_0^1 \frac{3}{4}y^2 dy + \int_1^{+\infty} \frac{3}{2}y^{-2} dy$$

$y < \frac{1}{x}$
 $y > x$

$$= \frac{1}{4} y^3 \Big|_0^1 + \frac{1}{2} y^{-2} \Big|_1^{+\infty}$$

$$= \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

$$E\left(\frac{1}{xy}\right) = \iint_D \frac{1}{xy} \cdot \frac{3}{2} x^2 y^2 d\sigma = \iint_D \frac{3}{2} x y d\sigma$$

$$= \int_0^1 \frac{3}{2} x^2 dx \int_x^{\frac{1}{x}} y dy = \int_0^1 \frac{3}{2} x^2 \left[\frac{1}{2x^2} - \frac{x^2}{2} \right] dx = \int_0^1 \left(\frac{3}{4} - \frac{3}{4} x^4 \right) dx$$

$$= \frac{3}{4} - \left(\frac{3}{4} \cdot \frac{x^5}{5} \right) \Big|_0^1 = \frac{3}{4} - \frac{3}{20} = \frac{9}{10}$$

300.4.9

4.9 设随机变量 X 与 Y 相互独立, 且 $X \sim N(1, 2), Y \sim N(1, 4)$, 则 $D(XY) = (C)$.

(A) 6

(B) 8

(C) 14

(D) 15

$$D(XY) = E(X^2 Y^2) - (E(XY))^2$$

$$= E(X^2) E(Y^2) - (E(X) E(Y))^2$$

$$= (2+1)(4+1) - 1 = 14 \quad C$$

300.4.10

4.10 设随机变量 X_1, X_2, X_3 相互独立, 且 $X_1 \sim B\left(4, \frac{1}{2}\right), X_2 \sim B\left(6, \frac{1}{3}\right), X_3 \sim B\left(6, \frac{1}{5}\right)$, 则

$$E[X_1(X_1 + X_2 - X_3)] = \underline{\quad\quad\quad}$$

$$E[X_1] + E(X_1)E(X_2) - E(X_1)E(X_3)$$

$$= 4 \cdot \frac{1}{2} + (4 \cdot \frac{1}{2})^2 + 2 \times 2 - 2 \times \frac{6}{5}$$

$$= 1 + 4 + 4 - \frac{12}{5} = \frac{33}{5}$$

300.4.11

4.11 设 X 为连续型随机变量, 方差存在, 则对任意常数 C 和 $\varepsilon > 0$, 必有 (C).

(A) $P(|X-C| \geq \varepsilon) = \frac{E(|X-C|)}{\varepsilon}$

(B) $P(|X-C| \geq \varepsilon) \geq \frac{E(|X-C|)}{\varepsilon}$

(C) $P(|X-C| \geq \varepsilon) \leq \frac{E(|X-C|)}{\varepsilon}$

(D) $P(|X-C| \geq \varepsilon) \leq \frac{DX}{\varepsilon^2}$

$$P\{|X-C| \geq \varepsilon\} = \int_{|x-c| \geq \varepsilon} f(x) dx$$

$$\leq \int_{|x-c| \geq \varepsilon} \frac{|x-c|}{\varepsilon} f(x) dx$$

$$\leq \int_{-\infty}^{+\infty} \frac{|x-c|}{\varepsilon} f(x) dx = \frac{1}{\varepsilon} E|X-C|$$

300.4.12

4.12 设随机变量 X 与 Y 相互独立且都服从参数为 p ($0 < p < 1$) 的几何分布, 令 $Z = X + Y$, 求:

(1) Z 的概率分布;

(2) X 与 Z 的相关系数.

$$(1) P_X(X=n) = (1-p)^{n-1} p$$

$$P_Y(Y=n) = (1-p)^{n-1} p$$

$$P_Z(Z=n) = \sum_{i=1}^{n-1} (1-p)^{i-1} p \cdot (1-p)^{n-i-1} p = \sum_{i=1}^{n-1} (1-p)^{n-2} p^2 = (n-1)(1-p)^{n-2} p^2$$

(2) $\text{Cov}(X, Z)$

$$= E(XZ) - E(X)E(Z)$$

$$= E[X(X+Y)] - E(X)E(X+Y)$$

$$= E(X^2) + E(XY) - E(X)^2 - E(X)E(Y)$$

$$= D(X) = \frac{1-p}{p^2}$$

$$p = \frac{\frac{1-p}{p^2}}{\sqrt{\frac{1-p}{p^2}} \cdot \sqrt{2 \cdot \frac{1-p}{p^2}}} = \frac{\sqrt{2}}{2}$$

300. 4.13

$$EY = 1 \frac{1}{3} + \frac{4}{3}$$

4.13 设二维离散型随机变量 (X, Y) 的概率分布为

X \ Y	0	1	2
	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
$\frac{1}{2}$	$\frac{1}{4}$	0	$\frac{1}{4}$
$\frac{1}{3}$	0	$\frac{1}{3}$	0
$\frac{1}{6}$	$\frac{1}{12}$	0	$\frac{1}{12}$

$$EX = \frac{2}{3}$$

求 $Cov(X, Y)$.

$$\begin{aligned} & Cov(X, Y) \\ &= E[(X-Y)Y] - E(X-Y)E(Y) \\ &= E[XY] - E(Y)^2 - E(X)E(Y) + E(Y)^2 \\ &= \frac{2}{3} - \frac{5}{3} - \frac{2}{3} + 1 = -\frac{2}{3} \end{aligned}$$

300. 4.14

4.14 设随机变量 $X_1, X_2, \dots, X_6$ 相互独立, 且同服从泊松分布 $P(\lambda_i)$ ($i=1, 2, \dots, 6$), 且 $Y_1 = X_1 + X_2 + X_3, Y_2 = X_4 + X_5 + X_6$, 则 (C).

(A) $Cov(X_1, Y_1) = \sum_{i=1}^3 \lambda_i$

(B) $Cov(X_1, Y_2) = \sum_{i=1}^6 \lambda_i$

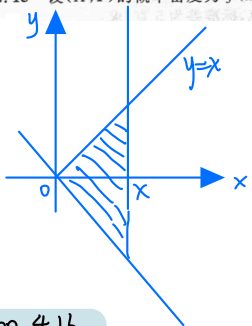
(C) $Cov(Y_1, Y_2) = 0$

(D) $Cov(Y_1, Y_2) = \sum_{i=1}^6 \lambda_i$

$$Cov(Y_1, Y_2) = E[Y_1 Y_2] - E[Y_1]E[Y_2]$$

300. 4.15

4.15 设 (X, Y) 的概率密度为 $f(x, y) = \begin{cases} 1, & 0 \leq |y| \leq x \leq 1, \\ 0, & \text{其他} \end{cases}$, 则 $Cov(X, Y) =$ D



$$\begin{aligned} Cov(X, Y) &= E[XY] - E[X]E[Y] \\ E[XY] &= \int_0^1 x dx \int_{-x}^x y dy = 0 \end{aligned}$$

300. 4.16

4.16 二维正态分布一般表示为 $N(\mu_1, \mu_2; \sigma_1^2, \sigma_2^2; \rho)$, 设 $(X, Y) \sim N(1, 4; 9, 0.5)$, 令 $Z = 2X - Y$, 则 Z 与 Y 的相关系数为

$$\begin{aligned} Cov(Z, Y) &= E[(2X-Y)Y] - E[2X-Y]E[Y] \\ &= E[2XY - Y^2] - 2E[X]E[Y] + E[Y]^2 \\ &= 2Cov(X, Y) - Cov(Y, Y) \\ &= -4 = -3 \end{aligned}$$

$$\sqrt{DZ} = \sqrt{4DX + DY - 2Cov(2X, Y)} = \sqrt{25 - 4 \times 2 \times 3} = \sqrt{13}$$

$$\rho = \frac{Cov(Z, Y)}{\sqrt{DZ} \sqrt{DY}} = \frac{-3}{\sqrt{13} \sqrt{9}} = -\frac{\sqrt{13}}{13}$$